XG Architecture Basics

RAN, Core, Security, and O-RAN

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CIS 4930/5930 Future Edge Networks
xinliulab.github.io/FSU-CIS4930-CIS5930-Future-Edge-Networks

What does the icon actually tell you?

Ericsson active 5G antennas

Turn off Wi-Fi for ten seconds.

What appears: LTE, 5G, 5G+, or 5G UW?

- **LTE:** 4G radio access.
- **5G:** 5G radio is available.
- **5G+:** AT&T mid- or high-band.
- **5G UW:** Verizon C-band or mmWave.

LTE = Long Term Evolution   mmWave = millimeter wave

Q: Does that icon describe the radio, the core, or the whole network?

A: Only the **radio** — the link between your phone and the tower — and even that is filtered through carrier branding. It says nothing about the **core**, the machinery behind the towers that knows who you are and decides where your traffic goes. And nothing about the speed you get

Somebody paid about $50 for that icon this month

What a line costs

- Entry unlimited: $45–$51
- Same price at 2 GB or 60 GB
- Hotspot only on some plans

What is not on the bill

- A price per gigabyte
- A speed you are owed
- Which radio you get

Entry unlimited, one line, autopay required, before taxes and fees. August 2026.

Q: If your bill does not change when your usage doubles, what are you buying?

A: Access, not data. You are buying the right to attach at all, which is why carriers compete on bundles instead of on megabits.

The bills are flat. The build is not.

What one line brings in

About $56.57 a month per postpaid phone line.

Down 0.3% in a year — and roughly flat for three.

What the build costs

$23–24 billion a year, guided through 2028.

Last quarter: $5.7 billion, against $4.9 billion a year earlier.

Both columns are AT&T, from its Q4 2025 and Q2 2026 SEC filings. Revenue per line is one carrier's; the pattern is the industry's.

ARPU = Average Revenue Per User   capex = money spent building things

Flat revenue per line, rising spend.
How long does that work?

Each generation made a new habit feel normal

1G	2G	3G	4G	5G
Mobile voice	Text and identity	Mobile Internet	All-IP broadband	Flexible systems

The app was visible.

The architecture underneath was not.

The first mobile call was basically a flex

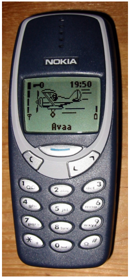
Martin Cooper with a 1973 DynaTAC prototype

Martin Cooper called his rival.

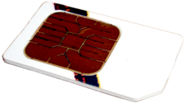
- Cooper was a Motorola engineer.
- In 1973 he called Joel Engel.
- Engel led AT&T's cellular team.
- AT&T was betting on car phones.
- Motorola bet on a handheld.
- Service reached users in the 1980s.

1G moved the telephone away from the wall.

The SIM card is the part that knows you



Nokia 3310



GSM SIM card

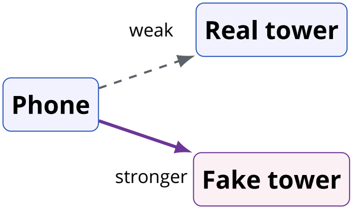
2G went digital.

- Digital voice and SMS
- Identity moved onto the SIM
- Mass-market roaming

SMS = Short Message Service   SIM = Subscriber Identity Module

Q: Which one is "you" to the network: the phone, the SIM, or the number?

A: The SIM. It stores the subscriber identity and the secret key used to authenticate you. The number is only a label the network maps onto that identity.



Three steps.

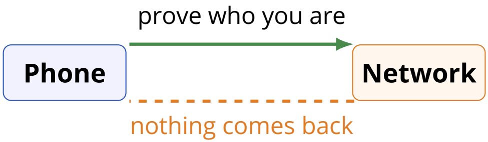
- The phone attaches to the strongest signal.
- The tower asks who it is.
- The phone answers — normally with a temporary ID.

Unless the tower insists on the permanent one.

IMSI = International Mobile Subscriber Identity, the permanent one

Equipment built to insist has a name: an IMSI catcher.

2G checked your ID. You never got to check theirs.



Q: The SIM proves the phone is real. What proved the tower was real?

A: In 2G, nothing — and the network also chose whether to encrypt at all. From 3G onward it must prove itself too, but the phone still has to pick a tower before it can check anything.

3G brought video calls before eye contact

Nokia 6680 commercial (YouTube)

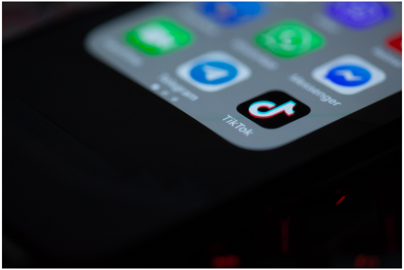
UMTS = Universal Mobile Telecommunications System
WCDMA = Wideband Code Division Multiple Access

Data became central.

- Web, email, multimedia
- UMTS/WCDMA radio access
- Packet data grew beside voice

3G made the phone a data device.
The core still ran voice and data as two separate worlds.

4G made buffering feel like a personal insult



TikTok app on phone

Broadband became the default.

- Streaming and short video
- Maps, ride sharing, live apps
- Voice moved to IP with VoLTE

VoLTE = Voice over LTE   IP = Internet Protocol

4G did not invent video. It made video normal.

The 5G icon can lie by omission

Trying AT&T's "5G E" (YouTube)

- It names a radio, not a speed.
- Branding sits on top of it.
- Spectrum and congestion decide.
- A 5G radio may use a 4G core.

Q: "Why is LTE faster than 5G here?"

A: A congested 5G cell on a narrow band can lose to a clean LTE cell. AT&T's old "5G E" icon was LTE-Advanced wearing a 5G name.

Paraphrased user question, 2026

You press Call. What happens next?

UMTS video call

Your phone has to find a tower.

The network has to recognize you.

Then it has to find the other end.

Q: Which part of the network handles each of those three jobs?

A: The RAN finds the tower and holds the radio link. The core recognizes you and decides what you may do. The core also reaches the other end, through the Internet or the telephone network.

Four worlds are hiding behind one tap

UE your device → **RAN** radio access → **Core** control + routing → **PSTN / Internet** destination

RAN

Turns radio waves
into managed links.

Core

Remembers who you are,
where you are,
and what you may do.

UE = User Equipment (your phone) RAN = Radio Access Network
PSTN = Public Switched Telephone Network, the traditional telephone network
3GPP = 3rd Generation Partnership Project, the body that writes these standards

Conceptual model based on 3GPP system architecture

The network still has the same six jobs

1. Find a cell
2. Request access
3. Prove identity
4. Get radio resources
5. Route voice or data
6. Move without dropping

The node names change.
The jobs are surprisingly persistent.

1G connected radio voice to the telephone network

Mobile phone → **Base station** → **MTSO** → **PSTN**

A circuit stays reserved
while the call is active.

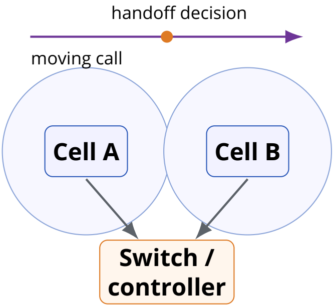
MTSO = Mobile Telephone Switching Office

PSTN = Public Switched Telephone Network

The path was a circuit.

- Analog radio in the cell
- MTSO linked the towers
- PSTN carried the call

A mobile call was still a telephone call.



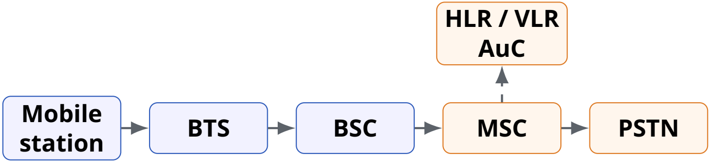
A moving car raises three questions:

- Who measures the signal?
- Who picks the next cell?
- Who keeps the call alive?

Q: What goes wrong if the handoff is late? If it is too early?

A: Late, the call drops while the old cell fades. Too early, the phone ping-pongs between cells and spends signaling on a move it did not need.

2G split the base station into radio and control

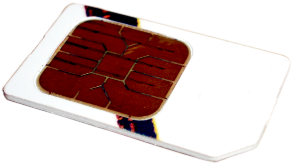


BTS	Base Transceiver Station
BSC	Base Station Controller
MSC	Mobile Switching Center
HLR / VLR	Home / Visitor Location Register
AuC	Authentication Center

Why split it? One BSC steers many BTSs. Radio hardware has to sit at every tower. Deciding to hand a call from one cell to the next has to see several towers at once. Those two jobs stopped fitting in one box.

Digital identities made roaming scalable.

The SIM became a network passport



GSM SIM card

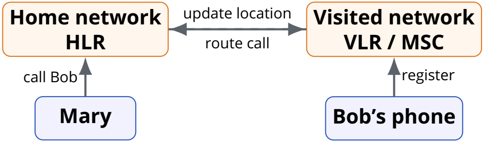
Home network
knows your subscription, number, and credentials.

Visited network
knows where to reach you right now.

Q: If you replace the phone but keep the SIM, what changed in the network?

A: Almost nothing. The network still authenticates the same subscriber record, so your number, your subscription, and your roaming state survive. Only the device identity and its radio capabilities changed.

A call can find you even when you leave home



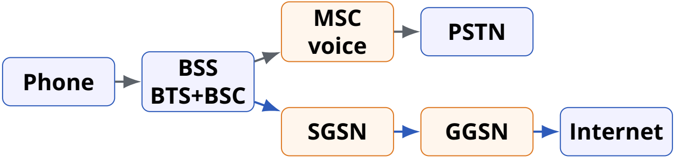
Bob visits another city.

1. Bob registers there.
2. The HLR is updated.
3. Mary's call is routed.

Mobility also needs a database.

2.5G: GPRS was an Internet sidecar

GPRS = General Packet Radio Service; packet data added onto 2G GSM.

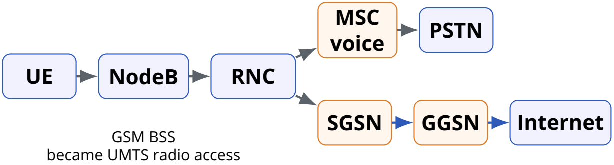


SGSN = Serving GPRS Support Node   GGSN = Gateway GPRS Support Node

- The 2G voice core stayed.
- Voice still used the MSC path.
- Packet data used SGSN and GGSN.

Packet data arrived next to voice, not instead of it.

3G renamed the radio side, but kept both worlds



NodeB = 3G radio node • RNC = Radio Network Controller

- BTS/BSC became NodeB/RNC in UMTS.
- Voice still reached the telephone network.
- Packet data still reached the Internet.

3GPP 3G overview

3G carried two networks in one trench coat

Voice

Circuit switched

MSC and the telephone world

Data

Packet switched

SGSN/GGSN and the Internet

Running both worlds at once worked. It was not simple.

Can you match the tower name to the generation?

1. BTS
2. NodeB
3. eNodeB
4. gNodeB

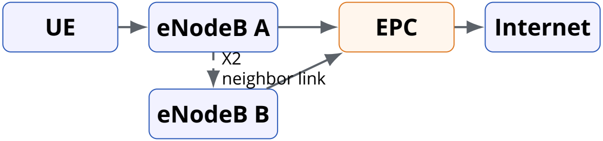
- 2G GSM
- 3G UMTS
- 4G LTE
- 5G NR

GSM = Global System for Mobile Communications NR = New Radio
The prefixes are generational: e = evolved (LTE), g = next generation (5G).

Q: Which generation removed the separate radio controller?

A: 4G. The eNodeB absorbed the scheduling and handover control that the RNC used to own, so LTE has no separate controller box in the main path.

4G flattened the RAN



No separate RNC sits in the main LTE path. EPC = Evolved Packet Core

- The eNodeB absorbed the radio control that lived in the RNC.
- Neighboring eNodeBs coordinate directly over X2.
- The core became the Evolved Packet Core.

Fewer boxes in the path.
More logic inside the base station.

4G asked: where did the voice path go?

Before LTE

UE → RAN → MSC → PSTN

circuit voice path

LTE answer

UE → eNodeB → EPC → IMS → Phone

voice as IP packets

VoLTE = Voice over LTE IMS = IP Multimedia Subsystem

- Early LTE often fell back to 2G/3G for calls.
- VoLTE kept the call on LTE as managed IP traffic.
- IMS controls the call; EPC carries the packet bearers.

Voice did not vanish.
The old circuit path vanished from LTE.

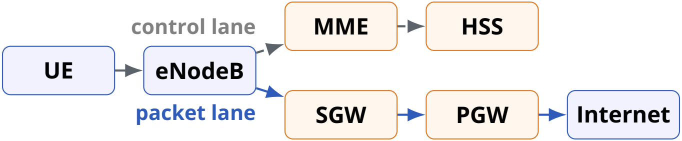
Early LTE	VoLTE
Data on LTE Voice falls back to 2G/3G	Voice as a managed IP service IMS provides call control

VoLTE = Voice over LTE   IMS = IP Multimedia Subsystem, the part of the core that sets up and controls calls

Q: Why was "all-IP" a bigger change than making the radio faster?

A: Speed is one number. All-IP changed the structure: one packet transport carries everything, and voice becomes an application on top of it instead of its own circuit-switched network.

EPC has two lanes: control and packets



dashed = control signaling • solid blue = user data

MME	Mobility Management Entity
HSS	Home Subscriber Server
SGW	Serving Gateway
PGW	Packet Data Network Gateway

The split is more abstract and higher-level:
control decides, user plane carries.

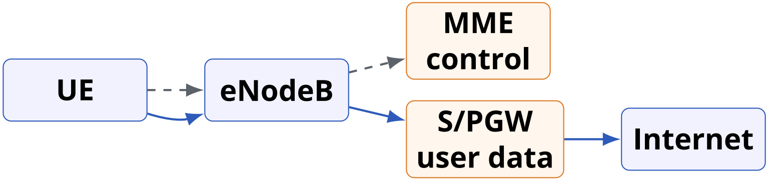
SGSN/GGSN jobs moved into EPC

2.5G/3G packet core	Main job	4G EPC home
SGSN	mobility + session	MME + SGW
GGSN	Internet gateway	PGW
Subscriber data	profile + permission	HSS

- Earlier packet boxes mixed decisions and traffic.
- LTE separated the story into clearer lanes.
- The gateway boxes still had some control logic, but the direction changed.

Same jobs, different boxes, clearer lanes.

The bigger pattern is decoupling



Small decisions should not scale like big traffic.

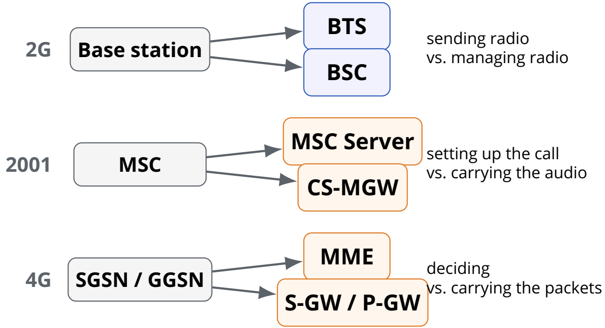
Control plane
small, slower decisions
identity, policy, mobility

User plane
large, fast traffic
voice, video, web packets

Voice/data was one split. Control/data is another.
O-RAN pushes the split into programmable control loops.

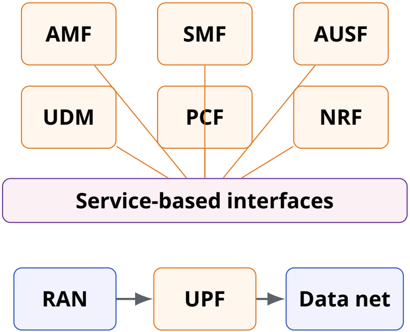
Decoupling lets each side evolve at its own speed.

This is not a coincidence anymore



One box becomes two along the seam between deciding and carrying.

5G made the core look more like a cloud platform



The SMF steers the UPF over the N4 interface.

Nokia 5G Core Service Based Architecture

AMF Access + Mobility

SMF Session Management

UPF User Plane Function

AUSF/UDM identity

PCF policy

NRF service registry

Core functions now discover and call each other as services.

Third time. Where did the MME go?

What it did

In 4G the MME was the one box that made control decisions: who may attach, where you are, which gateway serves you.

One box, every kind of decision.

What broke

Those decisions do not resemble each other.

Checking who you are happens once. Tracking where you are happens constantly. Setting up a session happens per app.

Same box, three completely different loads.

What replaced it

5G cut the control plane itself.

AMF handles access and mobility. SMF handles sessions. Each is a separate service that scales on its own.

AMF = Access and Mobility Management Function   SMF = Session Management Function

Same rule, one level deeper: now the control plane splits.

A 5G radio does not guarantee a 5G core

NSA = Non-Standalone   SA = Standalone   QoS = Quality of Service

NSA

LTE anchor → **4G EPC**

5G NR → **LTE anchor**

The phone shows 5G.
Control still runs through
LTE and the 4G core.

SA

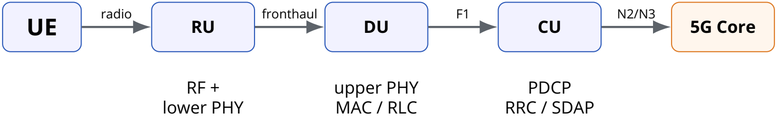
5G NR → **5G Core**

A 5G radio on a 5G core.
Slicing, edge, and new QoS
become possible.

The icon reports the radio.
The features come from the core.

AT&T: “Not all 5G is created equal,” 2025

The 5G base station can be split across locations



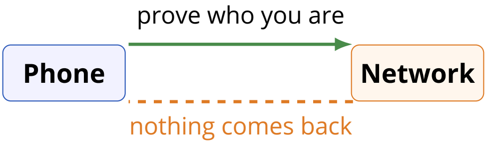
One gNodeB can be a system spread across a tower, a cabinet, and a data center.

RU / DU / CU = Radio / Distributed / Central Unit
PHY = physical layer MAC = Medium Access Control RLC = Radio Link Control
PDCP = Packet Data Convergence Protocol RRC = Radio Resource Control
SDAP = Service Data Adaptation Protocol

Q: Who owns each box, and who owns the interfaces between them?

A: In a classic RAN, one vendor owns all of it. Splitting the base station only helps if the seams between the pieces are specified well enough for a second vendor to plug in.

Security in the 2G Era



In 2G the phone proved itself to the network. The network proved nothing back, and the phone had already picked the loudest tower before anyone asked.

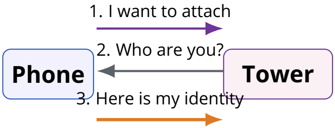
Every generation since has been trying to close that gap. Did they?

The same problem, four times

	The weakness	The answer
2G	Network never authenticates itself	—
3G	Identity still sent in the clear at first contact	Mutual authentication added
4G	Same first-contact exposure	Stronger keys, same opening move
5G	Phone must still pick a tower first	Identity encrypted before it is sent

Each generation closed a door.
None of them closed the doorway.

Why the first message is the hard one



Step 3 happens before the phone has verified anything about the tower.

The bootstrap problem.

To check a tower's credentials, the phone needs a channel.

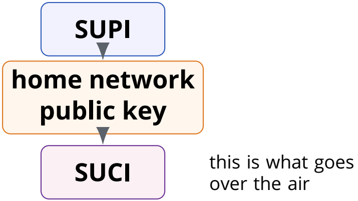
To get a channel, it has to talk to the tower.

Something has to go first.

IMSI = International Mobile Subscriber Identity

SUPI = Subscription Permanent Identifier

5G stopped sending the identity in the clear



SUCI = Subscription Concealed Identifier

Encrypt it first.

The permanent identity never leaves the phone unprotected.

What the tower receives is a concealed version that only the home network can open.

Q: So is the problem solved?

A: Partly. It closes the easy collection of permanent identities — but only where the operator has actually turned it on, and only if the phone stays on 5G.

Two ways it still leaks

Configuration

Concealment is a feature operators enable.

A network that has not turned it on gets none of the benefit, and the phone cannot tell from the outside.

Downgrade

Phones still support older generations so they keep working where 5G does not reach.

Push a phone down to a weaker generation and you are back to that generation's rules.

> Backward compatibility is a feature and an attack surface at once.

Downgrade behavior against 5G devices is demonstrated in the research literature. Treat it as a known capability, not as a claim about any particular network.

What would it take to notice?

- A real tower and a fake one look similar from one measurement.
- The signal that matters is a **pattern**: odd parameters, a cell that appears from nowhere, a push to an older generation.
- Seeing a pattern means watching many cells at once, over time.
- And the pattern changes faster than the standards do.

Where in the network could that detector even live?

Why this belongs in an architecture lecture

Every fix in this section was a change to **who talks to whom, in what order, and who is allowed to decide**.

That is not a security topic bolted onto an architecture topic.

It is the same topic.

Architecture decides what an attacker is allowed to try.

Can base-station security become an app?

New defense → **Vendor ticket** → **New image** → **Maintenance window**

You invent a defense against a new attack on the base station.

Q: Who installs it, who approves it, and what happens at 2:00 a.m.?

A: In a classic RAN the vendor answers all three. Your idea becomes a support ticket, then a firmware image, then a scheduled maintenance window — and nothing moves fast at 2:00 a.m.

> O-RAN's bet:
> some network behavior can ship as a tested application.

A traditional RAN often arrived as a sealed appliance

Telenor radio equipment rack

- One vendor shipped the box.
- One vendor owned the debugging.
- Upgrades followed its train.

Easy to blame one box. Hard to change one piece.

Four RAN models, three different questions

D-RAN process at site → **C-RAN** pool baseband → **vRAN** software functions → **O-RAN** open + intelligent

Virtualization
What runs where?

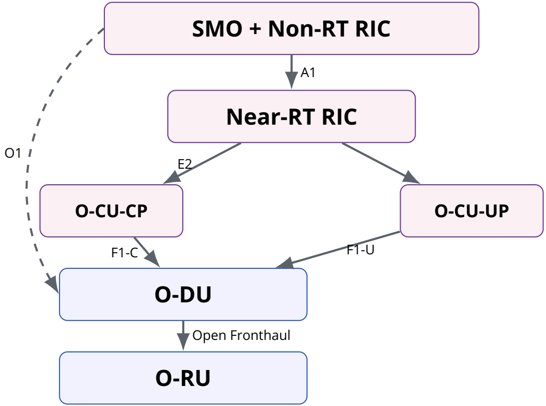
Disaggregation
Which functions split?

Open interfaces
Who can connect?

D-RAN = Distributed RAN C-RAN = Centralized (or Cloud) RAN
vRAN = Virtualized RAN O-RAN = Open RAN

Cloud software alone does not create an open, multi-vendor RAN.

O-RAN opens the seams, then adds control loops



O-RAN ALLIANCE architecture, redrawn from 2023 white paper

RAN chain
O-CU-CP / O-CU-UP
O-DU / O-RU

Control
SMO + Non-RT RIC
Near-RT RIC

Open seams
A1, E2, O1, Open Fronthaul

SMO = Service Management and Orchestration
RIC = RAN Intelligent Controller
Non-RT / Near-RT = non- and near-real-time
CP / UP = control / user plane

Every line on this diagram is a place a second vendor could plug in — and a place integration can fail

Open RAN is not the same as open-source RAN

Open RAN

Published interfaces

Disaggregated functions

Multi-vendor is possible

Open source

Source code is available

License permits reuse

Interface may stay closed

Q: If two boxes claim the same interface, will they work when plugged together?

A: Not automatically. A specification leaves options, versions, and timing choices open. Conformance and interoperability testing — plugfests, lab integration — is what finishes the job.

O-RAN ALLIANCE architecture and resources

RIC makes optimization an application

[Play: a real RIC demo (0:18–1:12)]

Non-RT RIC + rApp
policy and models, seconds and up

↓ A1

Near-RT RIC + xApp
faster decisions: roughly 10 ms–1 s

↓ E2

O-CU / O-DU / O-RU

Example: traffic steering
The rApp sets the policy.
The xApp reacts to load.
The RAN executes.

rApp / xApp = apps on the Non-RT / Near-RT RIC

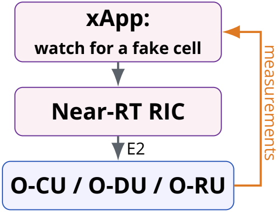
ITU, O-RAN / RIC demonstration

So: where does the fake-tower detector live?

Remember what detection needed:

- see many cells at once
- watch them over time
- update faster than the standard

That is the same shape as traffic steering. Which means it can be the same kind of thing.



Opening the seams turned “who would even build this?” into “who wants to write the app?”

This is the idea behind your recommended reading for today, 5G-Spector. It is a research prototype, not a deployed product.

Opening the RAN creates both options and work

Options

- More vendor choice
- Faster experiments
- Programmable control

Work

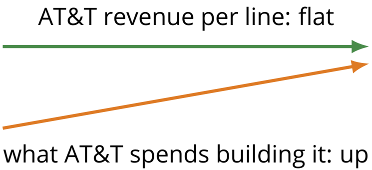
- Fronthaul and timing
- Integration testing
- Bigger security surface

Q: Would you rather debug one black box or five open interfaces?

A: Both are work, in different places. One box hides the failure and hands the whole fix to one vendor. Five interfaces make the failure visible, and make the integration yours.

O-RAN ALLIANCE overview (2026)

The question we opened with



Two lines going different directions.

How long can that last, and what happens at the end of it?

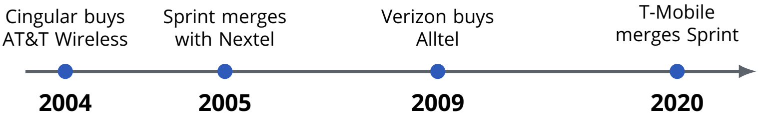
Roughly where a fifth of your bill goes

$17 billion	a year of network construction
$\div$	
126 million	postpaid connections
$\approx$ **$11**	per connection per month

Against a bill of roughly $50, that is about a fifth of what you pay going straight back into towers, spectrum, fiber, and power.

Deliberately crude: consolidated capex covers broadband and business too, and "connections" counts tablets and watches, not just phones. Verizon FY2025 actual capex and Q1 2025 connections.

Four national carriers became three



Years shown are when each deal **closed**. Brands lingered for years after, which is why you still hear some of these names.

This is a market that consolidates, not one that fragments.

So are they in trouble, or are they fine?

Fine

- Three players, high barriers
- Spectrum licenses are scarce
- Everyone needs a phone
- Steady, predictable revenue

In trouble

- Revenue per line is not growing
- Every generation costs billions
- Customers switch on price
- Debt from spectrum and mergers

Q: Which of these is a network architecture problem?

A: Most of them. Cheaper coverage, shared infrastructure, and software you can upgrade without replacing hardware are all architecture answers to a financial question. That is a large part of why O-RAN exists.

The competitor already in your pocket

Wi-Fi.

Every hour you spend on home or campus Wi-Fi is an hour the carrier does not have to build capacity for.

That is help — and it is also a ceiling on what they can charge for data you were never going to buy from them.

When did you last care which one you were on?

The answer is usually: when something broke.

A competitor you do not notice is still a competitor.

The competitor in orbit

Starlink direct-to-cell.

- Ordinary phones, no special hardware
- Started as a carrier partnership
- Then began looking like a rival

June 2026.

On the news that SpaceX might go direct to consumers, carrier shares fell sharply in a single session.

T-Mobile's CEO called the threat exaggerated.

As of late June 2026. This story is two months old and still moving — check what has happened since.

Who is right?

What a satellite can and cannot take

Satellites win

Where there are no towers and never will be: oceans, deserts, mountains, disasters.

Coverage is their whole argument.

Towers win

Where the people are. A stadium needs capacity in one small place, and capacity comes from cells close together.

One satellite covers a huge area — and everyone in it shares.

Coverage and capacity are different products. So far they are losing to each other on different ground.

The question to leave with

Every architecture change we walked through was somebody trying to serve more people for less money per person.

Splitting boxes. Moving voice onto IP. Turning the core into software.
Opening the RAN so you can buy the pieces separately.

If that pressure never stops, what does the network look like next?

Five generations in one low-resolution map

	Radio	RAN control	Core	Mode	Example
1G	Base station	MTSO	Analog core	Circuit	Voice in a car
2G	BTS	BSC	MSC + HLR/VLR	Circuit; GPRS packet	SMS, roaming
3G	NodeB	RNC	CS + PS cores	Both	Mobile web / video
4G	eNodeB	In the eNodeB	EPC, all-IP	Packet	Maps, apps, streaming
5G	gNodeB	CU (RRC), split CU/DU/RU	Service-based 5GC	Packet + QoS flows	Flexible services + edge

CS = circuit-switched PS = packet-switched QoS = Quality of Service
EPC = Evolved Packet Core 5GC = 5G Core

The radio name changes.
The deeper story is where control lives.

Follow the same action one last time

1G	2G	3G	4G	5G	O-RAN
circuit	identity	packet data	all-IP	modular core	open control

The packet still needs radio access,
identity + policy + routing,
and now sometimes programmable control.

Each generation relocates functions.
O-RAN changes who may control them.

Exit ticket: draw the boundaries

UE → **Radio system** → **Network services** → **PSTN / Internet**

1. Mark the RAN and Core boundary.
2. Add one O-RAN interface or controller.
3. Answer in one sentence:
   What changes next—radio, core, or control?

Appendix: the acronym decoder ring

Era	Node	Meaning / job
1G	MTSO	Mobile Telephone Switching Office; switching and mobility control
2G	BTS / BSC	Base Transceiver Station / Base Station Controller
2G/3G	MSC, HLR, VLR, AuC	Voice switch; home/visitor state; authentication
GPRS	SGSN / GGSN	Packet mobility + gateway to external packet networks
3G	NodeB / RNC	Radio node / Radio Network Controller
4G	eNodeB	LTE radio node; absorbs most RNC control
4G	MME, HSS, S-GW, P-GW	Mobility; subscriber data; serving and packet gateways
5G	gNodeB	5G radio node, often split into CU / DU / RU
5GC	AMF, SMF, UPF	Access/mobility; session control; packet forwarding
5GC	AUSF, UDM, PCF, NRF	Authentication; subscriber data; policy; service registry
O-RAN	SMO / RIC	Management/orchestration and intelligent control

Generic terms:
3GPP = 3rd Generation Partnership Project   IP = Internet Protocol
QoS = Quality of Service   PDU = Protocol Data Unit
CS / PS = circuit- / packet-switched   CP / UP = control / user plane

Appendix: interface map

Uu UE $\leftrightarrow$ radio: air

S1 / NG RAN $\leftrightarrow$ core

X2 / Xn neighbor handover

F1 O-CU $\leftrightarrow$ O-DU

Open FH O-DU $\leftrightarrow$ O-RU

O-RAN ALLIANCE architecture and resources

E2 Near-RT RIC $\leftrightarrow$ RAN

A1 Non-RT RIC $\leftrightarrow$ Near-RT RIC

O1 SMO $\leftrightarrow$ managed functions

O2 SMO $\leftrightarrow$ cloud platform

Interfaces expose coordination; they do not remove it.

Appendix: where the protocol stack is split

O-CU-CP **RRC + control-side PDCP**

E1

O-CU-UP **SDAP + user-side PDCP**

F1-U

O-DU **RLC + MAC + High-PHY**

Open Fronthaul

O-RU **Low-PHY + RF**

F1-C

Antenna $\longleftrightarrow$ air $\longleftrightarrow$ UE

A split is a placement decision.
Latency, bandwidth, compute, and ownership all move with it.

Appendix: handover changed owners, not purpose

Era	Main controller	What crosses the boundary?	What must survive?
1G	MTSO	Analog voice circuit	The conversation
2G	BSC / MSC	Radio channel + circuit state	Call and subscriber context
3G	RNC / core	Radio bearer; CS or PS state	Voice or packet session
4G	eNodeB + EPC	IP bearer and mobility context	All-IP session
5G	gNodeB + 5GC	PDU session, QoS flows, policy	Service continuity
O-RAN	Same 3GPP functions	RIC may influence parameters	Safe control under open APIs

Q: Which is harder to move—a radio channel, a packet tunnel, or its policy?

A: The channel has the hardest deadline; the policy is the hardest to keep consistent across cells, cores, and vendors.

Sources and further reading

- **3GPP TS 23.501**
  Authoritative 5G system architecture.
- **Nokia: 5G Core service-based architecture**
  Readable service-based-architecture role map.
- **O-RAN Architecture and Resources**
  Interfaces, RICs, and current specifications.
- **O-RAN ALLIANCE overview (2026)**
  Current official overview.
- **AT&T 5G Standalone announcement (2025)**
  Why a 5G icon can appear before a 5G core.
- **Course references**
  Local materials in `reference/` were used for background only, not reproduced as slide images.

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